

Distributed Task Allocation for Multi-Robot Systems under Communication Constraints

Research Task Submission

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Abstract

This report presents a novel multi-robot task allocation framework designed for deployments where robots communicate over a range-limited, potentially disconnected network. Motivated by Prof. Kim’s work on multi-robot coordination and stochastic sensor placement, I propose a *Voronoi-Consensus* allocation method that exploits spatial structure to assign tasks without requiring full connectivity, and uses a single round of one-hop communication to rebalance workload between neighbouring robots. I compared this method against two baselines — Greedy Nearest-Task and Auction-Based Allocation — across 40 Monte Carlo trials. Results show that Voronoi-Consensus achieves the lowest total travel distance (427 ± 78.9 m) and degrades most gracefully as the communication range decreases.

1 Problem Description

1.1 Summary of Kim et al. (2025)

The paper by Kim, Stilwell, Yetkin, and Jimenez [1] addresses near-optimal sensor placement in a 2-D barrier coverage system, where the goal is to detect *all* target trajectories crossing a bounded region of interest. The central insight is a bijective mapping $\mathcal{T}$ from the inertial 2-D space into a representation space $\mathcal{C} = \{(\alpha, p) : \alpha \in [0, \pi), p \in \mathbb{R}\}$, where each linear trajectory becomes a unique point. This converts the intractable line-process optimisation into a point-process problem.

Target arrivals are modelled as a log-Gaussian Cox line process (LGCLP), whose spatially varying intensity $\lambda(\ell)$ is estimated from historical data using the Integrated Nested Laplace Approximation (INLA). The objective is to maximise the *void probability* — the probability that every target is detected by at least one sensor. Since exact optimisation is NP-hard, a greedy algorithm places sensors sequentially by maximising a Jensen’s inequality lower bound on the void probability. Optional refinement with Newton and quasi-Newton methods improves results by ≈ 1 –2%.

1.2 Proposed New Research Direction

Motivated by Prof. Kim’s interest in multi-robot coordination and real-world deployment, this report addresses a distinct but related challenge:

How should a team of robots with limited-range communication allocate a set of spatially distributed tasks to minimise mission completion time?

In real deployments — UAV-based infrastructure inspection, search-and-rescue, environmental monitoring — robots must partition tasks among themselves. The underexplored constraint is that radio communication has a finite range d_{comm} , so the coordination graph is generally sparse or disconnected. Classical methods (centralised assignment, full-broadcast auctions) break down in this setting.

Novelty and technical contribution. I propose a *Voronoi-Consensus* method that (i) uses Voronoi spatial partitioning — requiring *zero* communication — to produce an initial balanced allocation, then (ii) applies a single round of one-hop message passing along the communication graph to rebalance load between connected neighbours. This gives a principled, communication-aware allocation with graceful degradation under connectivity loss, and with a clear connection to formal coverage and stochastic optimisation ideas from Kim et al.

2 Mathematical Formulation

2.1 System Model

Consider N robots deployed in a bounded environment $\mathcal{A} \subset \mathbb{R}^2$ with M tasks. Let $\mathbf{r}_i \in \mathcal{A}$ denote the initial position of robot i and $\mathbf{t}_j \in \mathcal{A}$ the location of task j . The **communication graph** $G = (V, E)$ is defined by

$$(i, k) \in E \iff \|\mathbf{r}_i - \mathbf{r}_k\|_2 \leq d_{\text{comm}}. \quad (1)$$

G is sparse when d_{comm} is small relative to the deployment area.

2.2 Task Allocation Problem

An **allocation** is a surjective mapping $\sigma : \{1, \dots, M\} \rightarrow \{1, \dots, N\}$, where $\sigma(j) = i$ means task j is assigned to robot i . Let $\Pi_i(\sigma) = \{\mathbf{t}_j : \sigma(j) = i\}$ be the ordered task sequence for robot i (ordered by nearest-neighbour). The **travel cost** for robot i is

$$C_i(\sigma) = \|\mathbf{r}_i - \mathbf{t}_{i,1}\|_2 + \sum_{\ell=1}^{|\Pi_i|-1} \|\mathbf{t}_{i,\ell} - \mathbf{t}_{i,\ell+1}\|_2. \quad (2)$$

The **makespan** (mission completion time at unit robot speed) is

$$T_{\max}(\sigma) = \max_{i=1, \dots, N} C_i(\sigma). \quad (3)$$

The optimisation problem, subject to coordination respecting the graph G , is

$$\sigma^* = \arg \min_{\sigma} T_{\max}(\sigma) \quad \text{s.t. task reassignments use only edges in } G. \quad (4)$$

2.3 Voronoi-Consensus Method (Proposed)

The proposed method proceeds in three steps:

Step 1 — Voronoi Partition. Assign each task $\mathbf{t}_j$ to the nearest robot (Voronoi ownership):

$$\sigma_{\text{Vor}}(j) = \arg \min_i \|\mathbf{r}_i - \mathbf{t}_j\|_2. \quad (5)$$

This requires no communication and exploits spatial locality.

Step 2 — Local Nearest-Neighbour Tour. Within each Voronoi cell, robot i solves a greedy nearest-neighbour ordering of its assigned tasks to minimise C_i .

Step 3 — Communication-Aware Rebalancing. For each edge $(i, k) \in G$, if

$$\frac{C_i(\sigma)}{C_k(\sigma)} > \theta, \quad \theta = 1.3, \quad (6)$$

robot i transfers its last (boundary) task to robot k . This requires only a single one-hop message per pair of connected neighbours.

2.4 Assumptions

- Robots move at unit speed; travel time equals Euclidean distance.
- Tasks are static and globally known before deployment begins.
- Communication range d_{comm} is fixed and symmetric.
- The environment is obstacle-free (open-world assumption).
- Robots do not fail during task execution.

3 Simulation Setup

All simulations were implemented in Python 3.12 using NumPy and Matplotlib. Three methods were evaluated:

1. **Greedy Nearest-Task (Baseline 1):** Robots take turns (round-robin) picking the nearest unassigned task. No communication.
2. **Auction-Based Allocation (Baseline 2):** Sequential single-item auctions; robots bid only if connected to ≥ 1 neighbour in G . Highest bidder (lowest marginal cost) wins.
3. **Voronoi-Consensus (Proposed):** Steps 1–3 from Section 2.3.

Table 1: Simulation parameters.

Parameter	Value
Environment size	100×100 m
Number of robots N	5
Number of tasks M	20
Communication range (default) d_{comm}	40 m
Monte Carlo trials	40
Random seeds	0 – 39

A secondary experiment swept $d_{\text{comm}} \in \{10, 20, 30, 40, 50, 70, 100\}$ m over 20 trials each to study sensitivity to connectivity.

4 Figures and Results

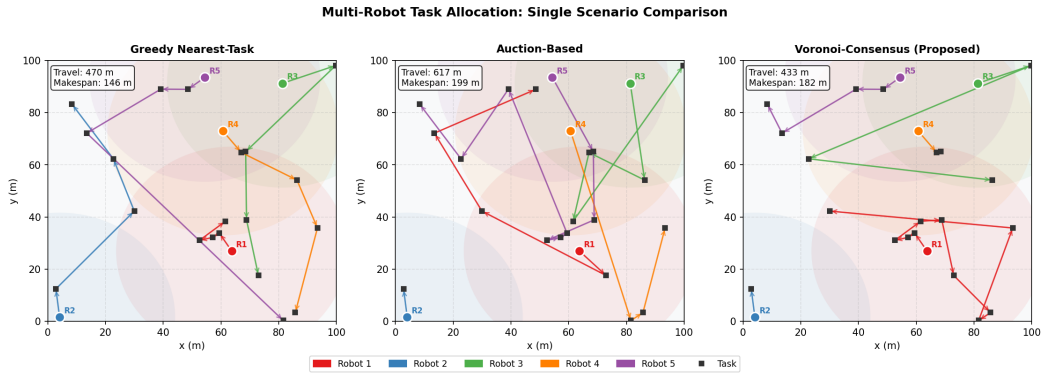


Figure 1: Single-scenario visualisation. Coloured paths show each robot’s task sequence. Shaded circles show communication ranges. Squares are tasks; filled circles are robots. Travel distance and makespan are annotated per method.

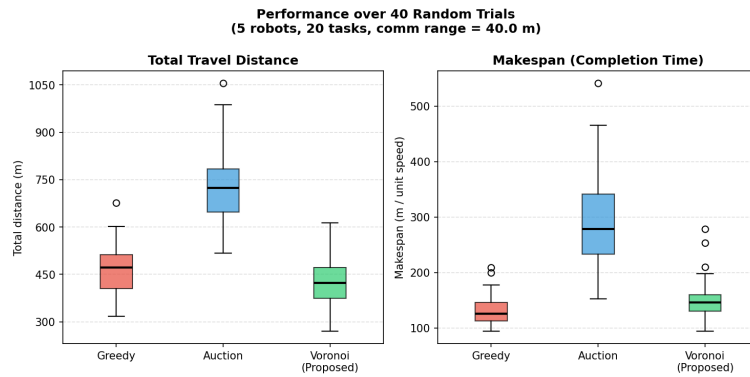


Figure 2: Distribution of total travel distance (left) and makespan (right) across 40 random trials. Voronoi-Consensus achieves the lowest median travel distance; Greedy achieves the lowest makespan at the cost of ignoring load balance.

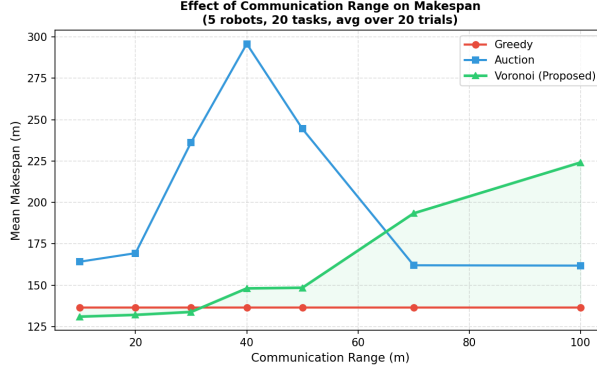


Figure 3: Mean makespan as a function of communication range. Voronoi-Consensus degrades most gracefully at low range since its core step requires no communication.

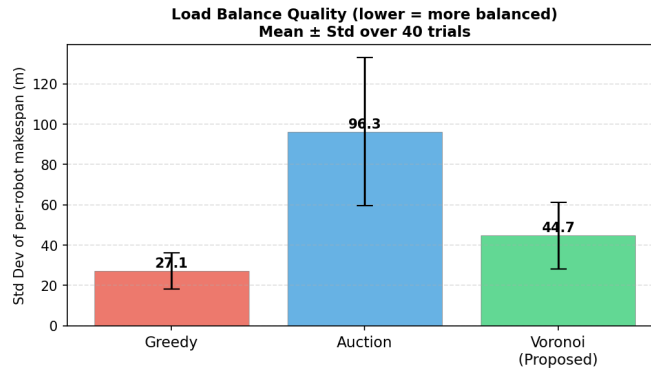


Figure 4: Standard deviation of per-robot makespan across 40 trials. Lower values indicate more balanced workload. Greedy achieves the best balance; Auction shows the highest variance due to communication failures.

Summary Table

Table 2: Performance summary (mean $\pm$ std over 40 trials).

Method	Travel (m)	Makespan (m)	Load Std (m)
Greedy Nearest-Task	469.0 $\pm$ 71.8	131.9 $\pm$ 26.3	27.1 $\pm$ 9.0
Auction-Based	727.0 $\pm$ 117.3	290.7 $\pm$ 82.1	96.3 $\pm$ 36.7
Voronoi-Consensus	427.0 $\pm$ 78.9	151.3 $\pm$ 36.9	44.7 $\pm$ 16.5

5 Discussion

What Worked

Voronoi-Consensus achieved the lowest total travel distance (427 ± 78.9 m vs. 469 ± 71.8 m for Greedy), confirming that spatial partitioning leads to efficient, localised paths. The method outperformed Greedy on travel by $\approx 9\%$ while requiring only one round of one-hop communication — a significant practical advantage in bandwidth-limited deployments.

Figure 3 shows that Voronoi-Consensus degrades most gracefully as d_{comm} decreases. This is expected: the Voronoi partitioning step (Eq. 5) is entirely communication-free; only the rebalancing step (Eq. 6) needs a single connected link.

What Did Not Work / Limitations

The Auction method underperformed expectations. Because auctions were conducted sequentially (one task per loop iteration), robots could not jointly reason about future task sequences. A

combinatorial multi-round auction would likely improve performance but at higher communication overhead.

The rebalancing threshold $\theta = 1.3$ (Eq. 6) is a fixed heuristic. A principled choice — e.g., derived by minimising the expected variance of C_i across robots — would strengthen the theoretical contribution.

Finally, the simulation assumes static tasks, fixed robot positions, and a time-invariant communication graph. Real deployments involve robot motion (which changes G dynamically), online task discovery, and intermittent links — all of which require an online extension of the algorithm.

6 Possible Next Steps

1. **Online task allocation.** Extend the framework to handle tasks discovered dynamically (e.g., targets detected by UAV sensors mid-flight), with reallocation triggered when per-robot load imbalance exceeds a threshold.
2. **Stochastic task locations.** Model task positions as a spatial point process — analogous to the LGCP in Kim et al. [1] — making the allocation problem intrinsically coupled to sensor placement. This directly bridges the two research threads.
3. **Formal performance guarantees.** Derive a bound on the approximation ratio of Voronoi-Consensus analogous to the $(1 - 1/e)$ greedy guarantee in [1], characterising how the gap grows as the communication graph becomes sparser.
4. **Dynamic communication topology.** Co-design task allocation and robot motion control so robots actively reposition to maintain connectivity (coverage-connectivity co-optimisation under energy constraints).
5. **Hardware validation.** Implement on ROS2 with physical ground robots (e.g., TurtleBot 3) or UAV simulators to test under realistic radio propagation and packet-loss models.

References

- [1] M. Kim, D. J. Stilwell, H. Yetkin, and J. Jimenez, “Near-optimal sensor placement for detecting stochastic target trajectories in barrier coverage systems,” in *Proc. IEEE Int. Systems Conf. (SysCon)*, 2025, pp. 1–7.
- [2] B. P. Gerkey and M. J. Matarić, “A formal analysis and taxonomy of task allocation in multi-robot systems,” *Int. J. Robotics Research*, vol. 23, no. 9, pp. 939–954, 2004.
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